Revision log for "On the unitary dual of $\overline{SL(2,F)}$, F p-adic"

Draft 1 (November 9, 2026) to Draft 2 (November 13, 2026)

Prepared by Claude (Anthropic) for D. Joyner

November 13, 2026

Contents

1 Files **1**

2 AI usage **2**

3 Original results and results new in the revision **2**
3.1 Results of the 1999 manuscript 2
3.2 New in Draft 1 (Sol-ChatGPT) 3
3.3 New in Draft 2 (Claude) 3

4 Changes from Draft 1 to Draft 2 **4**
4.1 Style-guide conformance 4
4.2 Mathematical changes and corrections 5
4.3 Removed material 5
4.4 Bibliography corrections 5
4.5 Citation keys 6

5 Sources consulted and not consulted **6**

6 Verification of the mathematics **7**

7 The verification script **8**

8 Concordance of numbering **8**

1 Files

- `sl2bar-unitary-dual-2026.tex`: Draft 2 of the paper.
- `revision-log.tex`: this log.

- `scripts/kubota_torus_check.py`: exact-arithmetic checks of the cocycle computations (Section 7).
- Inputs:
  - the 1999 manuscript `sph_corr5b.tex`, dated 7-21-99,
  - Draft 1, in a file whose name ends in `metaplectic_cover_2026c.tex`,
  - the style guide, Draft 2 of November 11, 2026, in `joyner-math-paper-style-guide.md`.

Draft 2 compiles with `pdflatex` with no errors, no undefined references and no overfull boxes. It was compiled here with `mathptmx`, because `newtx` is not installed in the sandbox. The preamble loads `newtxtext` and `newtxmath` when they are installed, so a local build will use them. It has not been test-compiled with `newtx`.

2 AI usage

- **Argon-Gemini.** Refereed the 1999 manuscript, suggested corrections, added a glossary and produced a first revised manuscript. A later revision attempt was not completed. (Source: the abstract and acknowledgments of Draft 1.)
- **Sol-ChatGPT (OpenAI).** Wrote an independent referee report and prepared Draft 1, the corrected LaTeX edition. (Source: Draft 1.)
- **Claude (Anthropic), this draft.** Rewrote Draft 1 to follow the style guide. Checked every mathematical claim of Draft 1 independently (Section 6). Supplied complete proofs where Draft 1 gave sketches. Wrote the Background section, all new examples, and the expanded literature review. Checked bibliographic data by web search. Wrote the verification script and this log.

Claude did not see the Argon-Gemini manuscript or either referee report. Claude's knowledge of them comes only from Draft 1.

3 Original results and results new in the revision

Rule 6.5 asks for this separation.

3.1 Results of the 1999 manuscript

1. Vanishing lemma for orbital integrals off the dth powers. Kept (Lemma 6.1), with the correct quotient.
2. Surjectivity of the split orbital-integral map (sketched in 1999, with the final step referred to Shalika germs). Kept and now fully proved (Theorem 6.4).
3. Description of maximal abelian subgroups and genuine representations of $\widetilde{A}$. The 1999 "2-1 correspondence" $\Pi(A_n) \to \Pi(\widetilde{A})_{\mathrm{gen}}$ is wrong as stated, see Section 4.3. Replaced by Theorem 5.6.

4. Character formula for principal series, stated with $\operatorname{tr}\chi(a)+\operatorname{tr}(\chi(a)^{-1})$ and with proof omitted. Corrected to Weyl conjugation and proved (Theorem 8.3).

5. Character identity $\operatorname{tr}\overline{\pi}(\bar{f})=\operatorname{tr}\pi(f)$ for $\gcd(p,2n)=1$ and all n. Correct only for n odd or $4 \mid n$, with the normalization of Theorem 9.4.

6. Unitarity of principal series, complementary series for $0<s<1/n$, the special representation and the "trash" representation at $s=1/n$. Not proved. Recorded as cited material in Remarks 2026.2–2026.5.

3.2 New in Draft 1 (Sol-ChatGPT)

The correct center Z_A and the index d^2. The Stone–von Neumann parametrization by genuine characters of Z_A. The distinction between Weyl conjugation and inversion. The restriction to n odd or $4 \mid n$ and the power map ν (Lemma 9.1). Explicit Haar normalizations and the d versus d^2 distinction. Removal or demotion of the unproved unitary-dual claims.

3.3 New in Draft 2 (Claude)

- Lemma 3.2 (cocycle values, uniqueness of the splitting of N, centrality of the preimage of $\pm I$). New, with proof.
- Lemma 5.1 (exact commutator $(a,b)_n^{-2}$). Draft 1 had $(a,b)_n^{\pm 2}$.
- Lemma 5.4 (Weyl conjugation on $\widetilde{A}$). Draft 1 stated (b) inside the proof of Lemma 8.1. Parts (a), (c), (d) are new.
- Lemma 6.3 (constant term). Draft 1 stated the formula without proof.
- Theorem 6.4. Draft 1 gave a sketch. Step 1 (submersion argument, fiber description, explicit preimage) and Step 2 (explicit functions near $\pm I$ and their transforms) are written out.
- Proposition 7.1. Draft 1 cited Moen for irreducibility. Draft 2 proves (a) directly and (b) from Frobenius reciprocity and the geometric lemma, which are cited (Remark 2026.2).
- Lemma 7.2 (spherical vectors). New.
- Lemma 8.1 (trace of an induced representation by the kernel method). New. Corollary 8.2 now follows from it directly.
- Theorem 8.3. Draft 1 gave a sketch citing [G, F, FK]. Draft 2 proves well-definedness, the trace identity via the Weyl integration formula, and local integrability.
- Section 3 (Background) and Examples 3.1, 3.3–3.7, 5.5, 6.2, 6.6, 7.3, 8.5, 9.3. New. Examples 5.7, 5.8, 9.5 are from Draft 1, revised.

4 Changes from Draft 1 to Draft 2

4.1 Style-guide conformance

- **1.1.** Twelve-point Times (`newtx`, with `mathptmx` fallback), US Letter, one-inch margins. Draft 1 used Latin Modern.
- **1.2, 1.3.** Compiles cleanly. Two overfull boxes were removed by rewording.
- **1.5.** All cross-references use `\ref` and all citations use `\cite`. Section titles with math use `\texorpdfstring`.
- **1.7.** Citation keys changed to author letters plus year (table in Section 4.5). Exception: Moen's undated preprint has key `Mo`, because its year is unknown.
- **1.8.** Operators declared with `\DeclareMathOperator`.
- **1.9.** Every Draft 1 label is kept, except `eq:d`, which was a duplicate definition of d. The macro `\ht` was renamed `\hl`, since `\ht` is a TEX primitive.
- **2.** Order: abstract, introduction, literature review, background, notation, body, summary and open questions, glossary, acknowledgments, references. Draft 1 had the notation table before the introduction and no background section.
- **2.3.** Literature review split at 1999. Works added: [Mc09], [Li12], [Lu20], [Sz24], [Ne86], [Si79].
- **2.4, 3.3.** Each background subsection and each section has at least one example.
- **3.2.** Every section and subsection opens with an orienting paragraph.
- **3.8.** Prose has no semicolons. Sentences were shortened.
- **5.1–5.5.** Abstract opens with the AI disclosure. Email added to the author line. Acknowledgments name each AI and its role, say that AI review is not certification, and point to this log.
- **6.2.** Title restored to the 1999 title. Draft 1 had "On the unitary dual of the metaplectic cover of $\mathrm{SL}(2,F)$, F p-adic". Subtitle "Draft 2. Corrected and expanded version". Date line "Original draft: 1999. Corrected and expanded draft: 2026."
- **6.3.** Note on the original manuscript in a `\thanks` footnote.
- **6.4.** Revision history removed from the body (the Draft 1 acknowledgments paragraph on revision history, Draft 1 Remark 4.6, and the phrase "the earlier draft" in Section 6.3 of Draft 1).
- **6.6.** Statements not fully verified are labeled Remark 2026.1–2026.5.

[DECIDE] items, defaults used. Document class: kept `article` from Draft 1. Times package: `newtx` with fallback. Glossary at the end, before the acknowledgments. Author line: Draft 1 listed Argon-Gemini and Sol-ChatGPT after "with". Claude was added to that list because it supplied new proofs. Remove it if you prefer.

4.2 Mathematical changes and corrections

1. **Example 5.7 (double cover).** Draft 1 said "This does not imply that the double cover splits over A." That is true as a statement of logic, but it suggests that the cover does not split over A. In the tame case it does split as an abstract group: $\varpi^k u \mapsto \tilde{h}(\varpi)^k \tilde{h}(u)$ is a homomorphism when $n = 2$, since $\widetilde{A}$ is abelian and the cocycle is trivial on units. The splitting is not canonical. Replaced by the true statement that $t \mapsto \tilde{h}(t)$ is not a homomorphism.

2. **Theorem 8.3 hypothesis.** Draft 1 assumed $\gcd(p, n) = 1$. Draft 2 assumes $\gcd(p, 2n) = 1$, because the proof uses the splitting of K through Corollary 8.2. The formula itself is expected to hold for $p \nmid n$, but that is not proved here.

3. **Intertwining operator (Section 7.4).** Draft 1 wrote $T_s\varphi(g, a)$ in a model with an extra variable a that was never defined. Replaced by the standard operator $\varphi \mapsto \int_F \varphi(\tilde{w}^{-1}(u(x), 1)g)\, dx$, with the matrix J_s described in words.

4. **Example 9.5 (degree three).** Draft 1 asserted the value of the constant term of e_K. Draft 2 derives it (Example 6.6) and checks the answer against Lemma 7.2 and the linear side (Examples 3.7, 9.3).

5. **Proposition 7.1.** Draft 1 also asserted temperedness. That is now in Remark 2026.2 as unverified for covers.

6. **Remark 2026.4.** Records exactly what the 1999 manuscript asserted about complementary series and the representations at $s = 1/n$, and why it is not proved.

4.3 Removed material

Draft 1 Remark 4.6 ("Correction of an earlier multiplicity claim") compared with the 1999 text, so it moved here. Its content: the 1999 lemma claimed a 2-1 map $\Pi(A_n) \to \Pi(\widetilde{A})_{\text{gen}}$ through characters η_j that differ on $C \setminus A_n$. For odd n, $A_n = C$, so the two choices coincide and the claim fails. The correct invariant is the genuine central character on Z_A.

4.4 Bibliography corrections

Ma84a, Ma84b : year 1987 corrected to 1984, and the pages of part II corrected from 297–305 to 297–306.

Mo88 : title corrected to "Irreducibility of unitary principal series for covering groups of $SL(2, k)$". Author C. H. Moen.

Mo : Draft 1 described it as a 1979 University of Chicago Ph.D. thesis. We could not confirm this, so the entry reverts to the 1999 description, "preprint". Draft 1 also referred to Moen as "her". The pronoun was removed.

Ta17 : author's name S. Tang (Shiang Tang). The Draft 1 summary was corrected. The paper computes Plancherel measures by a comparison method and constructs genuine central characters of the torus in the simply laced case.

GS75 : title "…for the metaplectic group" (Draft 1 had "on").

Jo99 : volume and pages 73 (1999) 332–340 confirmed. Older citations give "273, 1–9", which is wrong.

4.5 Citation keys

Draft 1	Draft 2	Draft 1	Draft 2
Ar	Ar80	K	Ku67
BZ1, BZ2	BZ76, BZ77	KP	KP84
F	Fl80	M1, M2	Ma84a, Ma84b
FK	FK86	Mo1, Mo2	Mo, Mo88
G	Ge76	Sch	Sc98
GS	GS75	Szpruch19	Sz19
GoldSz	GS16	Tang	Ta17
J1, J2, J3, J4	Jo88, Jo98, Jo99, Jo01	(new)	BD84, Li12, Lu20, Mc09, Ne86, Si79, Sz24

5 Sources consulted and not consulted

Rule 4.3. Level 2 means checked against a source actually consulted. Here that source was usually an abstract, a publisher page or an arXiv listing, found by web search on November 13, 2026. No full text was read in this revision.

Key	What was checked
Ar80	Level 2: bibliographic data and the introduction (irreducible off $\mathrm{Re}\, s = 0$ except at $s = \pm 1/3$).
BD84	Not consulted. The claim that the arguments extend to finite central extensions comes from the 1999 manuscript.
BZ76, BZ77	Not consulted. Precise theorem numbers for the geometric lemma not given.
Fl80, FK86, KP84	Bibliographic data only (FK86 and KP84 confirmed in another paper's reference list).
Ge76, GS75	Bibliographic data only. The endpoint $s = 1/2$ statement in Remark 2026.4 is from memory (Level 3).
GS16	Level 2: journal data, the assumption $p \nmid n$, and Theorem 5.2 (reducible genuine principal series from unitary data), from the arXiv introduction.
Jo88	Not re-checked. Volume 61 agrees with Draft 1 and with the October 2026 corrected edition of Jo01. The 1999 manuscript has 66.
Jo98, Jo01, Jo99	Level 2: journal data. Jo01's reference list includes the 1999 manuscript as a preprint.
Ku67	Not consulted. Formula (3.2) for the splitting of K was spot-checked by script (Remark 2026.1).

Key	What was checked
Li12	Theorem numbers 2.6.4, 5.6.3, 5.8.10 are taken from the October 2026 review of Jo01, when the PDF was read. Not re-checked here.
Lu20	Level 2: abstract.
Mc09	Level 2: arXiv abstract and introduction (Stone–von Neumann for the torus, Satake isomorphism). The published version (believed to be in Progress in Math. 300) was not confirmed, so the arXiv version is cited.
Mo	Not found. See above.
Mo88	Level 2: publisher page and abstract (p odd).
Ne86	Not consulted. Standard properties of the Hilbert symbol.
Sc98	Level 2: title page of the thesis.
Si79	Not consulted. The 1999 manuscript cites p. 198 for the Weyl integration formula. The page was not checked, so no page is given.
Sz19	Level 2: journal data and abstract ($p \mid n$ allowed, $4 \nmid n$ assumed).
Sz24	Level 2: arXiv abstract and introduction (reducible unitary genuine principal series parametrized by nontrivial quadratic characters, odd n, two summands).
Ta17	Level 2: arXiv abstract.

6 Verification of the mathematics

Level 1 means an independent derivation by Claude in this revision.

- Level 1: Lemmas 3.2, 5.1, 5.2, 5.4, 6.1, 6.3, 7.2, 8.1, 9.1, Theorems 5.6, 6.4, 8.3, 9.4, Corollary 8.2, Proposition 7.1(a), and all examples. The proofs in the paper are the derivations.
- Level 1 modulo citations: Proposition 7.1(b) (Frobenius reciprocity and the geometric lemma for covers). Theorem 8.3 (Weyl integration formula for $SL(2,F)$). Theorem 6.4 Step 1 (p-adic inverse function theorem). Section 3.2 (formula (3.2) for the splitting of K).
- Level 3 or cited only: Remarks 2026.2–2026.5.

Points a human referee should check first.

1. Formula (3.2) for the splitting of K, and that $\kappa = 1$ on upper triangular matrices. Lemma 7.2 and Examples 6.6 and 9.5 depend on it.
2. The Haar normalization (3.4), and that the quotient measure on $\widetilde{A}\backslash\widetilde{G}$ is $du\,dk$.
3. Step 2 of the proof of Theorem 6.4.
4. The kernel argument in Lemma 8.1.

7 The verification script

`scripts/kubota_torus_check.py` models $F = \mathbb{Q}_p$ with $n \mid p-1$. It represents elements of $F^\times$ by rationals and uses the tame formula (H5). All checks are exact. For

$$(p,n) \in \{(7,3),(13,3),(5,4),(13,4),(7,6),(13,6),(3,2),(7,2)\}$$

it checks:

1. the cocycle identity for β on random elements of $SL(2,\mathbb{Q})$, which validates the implementation;
2. $\beta(h(a),h(b)) = (b,a)_n$ (Lemma 3.2(a));
3. the commutator $(a,b)_n^{-2}$ (Lemma 5.1);
4. $\beta(w,w^{-1}) = 1$ and $\tilde{w}^{-1}\tilde{h}(t)\tilde{w} = \tilde{h}(t^{-1})$ (Lemma 5.4(a), (b));
5. $\tilde{h}(t)^{-1} = (t,-1)_n\tilde{h}(t^{-1})$ (Lemma 5.4(c));
6. the center is $(F^\times)^d$ with index d^2 (Lemma 5.2);
7. ν is a homomorphism exactly when $n \mid d^2$ (Lemma 9.1 and Remark 9.6). It fails for $n = 2$ and $n = 6$, as expected;
8. $u \mapsto (\varpi,u)_n^2$ has order d on units (Lemma 7.2);
9. the splitting identity for κ on random elements of $SL(2,\mathbb{Z}_p)$ (Remark 2026.1).

It also checks $(5,-1)_4 = -1$ (Example 3.1). All checks pass, and the script exits with status 0. Run time is about one second. The name ends in `_check.py`, so the repository's `run_checks.py` runs it with the other checks.

8 Concordance of numbering

Draft 1 numbers on the left, Draft 2 numbers on the right. Unlabeled Draft 1 items are matched by content.

Item	Draft 1	Draft 2
Sections: intro, literature	1, 2	1, 2
Covering groups and conventions	3	3.2–3.4 (Background)
Table of notation	unnumbered, before §1	4
Covering torus, orbital integrals	4, 5	5, 6
Parabolic induction	6	7
Characters, transfer	7, 8	8, 9
Conclusions, glossary	9, 10	10, 11
Lemma radical	4.1	5.2
Remark on centralizers	4.2	5.3

Item	Draft 1	Draft 2
Theorem torus	4.3	5.6
Example 2 and 3	4.4	5.7
Example 4 and 6	4.5	5.8
Remark on multiplicity claim	4.6	removed (this log, §4.3)
Lemma vanishing	5.1	6.1
Theorem orbital	5.2	6.4
Remark Satake	5.3	6.7
Proposition tempered	6.1	7.1
Remark reducibility	6.2	Remark 2026.3
Theorem character	7.1	8.3
Caution	7.2	8.4
Corollary trace	7.3	8.2
Lemma norm	8.1	9.1
Definition matching	8.2	9.2
Theorem transfer	8.3	9.4
Example degree three	8.4	9.5
Remark Weil	8.5	9.6
Remark spherical	8.6	9.7
Equations (3.1), (3.2)	kubota, genuine	(3.1), (3.3)
Equations (3.3), (3.4)	D, δ_B	(3.5), (3.6)
Equation (3.5)	d	removed (defined in (5.2))
Equations (4.1)–(4.5)		(5.1)–(5.5)
Equations (5.1)–(5.5)	OZ, Phi, index, Abel, surj	(6.1), (6.2), (6.3), (6.5), (6.6)
Equations (6.1), (6.2)		(7.1), (7.2)
Equations (7.1)–(7.4)	char, char-Z, trace	(8.3), (8.4), (8.1), (8.2)
Equations (8.1)–(8.7)		(9.1)–(9.7)